

3-Sex discrimination case study (1974)

MATH 213

Front Window Side Table

Is it possible this difference has arisen by random chance alone due to the variability associated with randomly assigning the supervisors to the “male file” and “female file” groups?

We will try to answer this question by considering two competing claims:

Null hypothesis, H_0 : Each supervisor would make the same decision (whether to promote or not) regardless of the gender listed on the file. The difference in promotion proportions of 29.2% between the groups is due to the variability coming from the random assignment of supervisors to the “male file” and “female file” groups.

Alternative hypothesis, H_a : There is at least one supervisor who would decide to promote based on a male file but to not promote based on a female file.

In the study, 35 out of the 48 supervisors decided to promote. One way to examine this situation is to assume that 35 of the supervisors were going to promote regardless of the gender listed on the candidate's file. The key question is **how unlikely is it for the random assignment process alone to randomly place 21 (or more) of these 35 “promoting supervisors” into the “male file” group?**

1. Take a set of 48 playing cards, 35 of which are blue, each representing a supervisor who decides to “promote”, and 13 of which are red, each representing a supervisor who decides to “not promote”. Count the cards to make sure this is what you have.
2. Shuffle the cards well. Then shuffle them some more. And a whole bunch more to make sure they are well mixed.
3. After shuffling well, deal out two groups: 24 of the supervisors to the “male file” group and 24 of the supervisors to the “female file” group.
4. Count how many “promoting supervisors” (blue cards) are in each group, filling out the table below.

Simulation Rep 1	Male File Group	Female File Group	Total
Promote	15	20	35 (blue cards)
Not promote	9	4	13 (red cards)
	24	24	48

5. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$15/24 = 0.625$$

6. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$20/24 = 0.833$$

7. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.625 - 0.833 = -.208$$

8. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 1.

9. Shuffle the cards extremely well again and repeat this process two more times to produce a difference in proportions for two more simulations. Use the tables and questions on this and the next slide to organize your work.

Simulation Rep 2	Male File Group	Female File Group	Total
Promote	15	20	35 (blue cards)
Not promote	9	4	13 (red cards)
	24	24	48

9a. Compute p_{male} = the proportion of supervisors in the male file group who promoted

$$\mathbf{15/24 = 0.625}$$

9b. Compute p_{female} = the proportion of supervisors in the female file group who promoted

$$\mathbf{20/24 = 0.833}$$

9c. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$\mathbf{0.625 - 0.833 = -0.208}$$

9d. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 2.

Simulation Rep 3	Male File Group	Female File Group	Total
Promote	17	18	35 (blue cards)
Not promote	7	6	13 (red cards)
	24	24	48

9e. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$17/24 = 0.708$$

9f. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$18/24 = 0.75$$

9g. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.708 - 0.75 = -0.042$$

9h. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 3.

We will analyze your work together to see how unusual the proportion of 0.29 is if the managers really would make the same decision regardless of the gender marked on the file.

After this, we will automate our simulation. You will need to go to this link:

<https://www.lock5stat.com/StatKey/index.html>

Next time, we will learn how to do this “randomization test” in R.

Front Door Side Table

Is it possible this difference has arisen by random chance alone due to the variability associated with randomly assigning the supervisors to the “male file” and “female file” groups?

We will try to answer this question by considering two competing claims:

Null hypothesis, H_0 : Each supervisor would make the same decision (whether to promote or not) regardless of the gender listed on the file. The difference in promotion proportions of 29.2% between the groups is due to the variability coming from the random assignment of supervisors to the “male file” and “female file” groups.

Alternative hypothesis, H_a : There is at least one supervisor who would decide to promote based on a male file but to not promote based on a female file.

In the study, 35 out of the 48 supervisors decided to promote. One way to examine this situation is to assume that 35 of the supervisors were going to promote regardless of the gender listed on the candidate's file. The key question is **how unlikely is it for the random assignment process alone to randomly place 21 (or more) of these 35 “promoting supervisors” into the “male file” group?**

1. Take a set of 48 playing cards, 35 of which are blue, each representing a supervisor who decides to “promote”, and 13 of which are red, each representing a supervisor who decides to “not promote”. Count the cards to make sure this is what you have.
2. Shuffle the cards well. Then shuffle them some more. And a whole bunch more to make sure they are well mixed.
3. After shuffling well, deal out two groups: 24 of the supervisors to the “male file” group and 24 of the supervisors to the “female file” group.
4. Count how many “promoting supervisors” (blue cards) are in each group, filling out the table below.

Simulation Rep 1	Male File Group	Female File Group	Total
Promote			35 (blue cards)
Not promote			13 (red cards)
	24	24	48

5. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)
6. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)
7. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).
8. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 1.

9. Shuffle the cards extremely well again and repeat this process two more times to produce a difference in proportions for two more simulations. Use the tables and questions on this and the next slide to organize your work.

Simulation Rep 2	Male File Group	Female File Group	Total
Promote			35 (blue cards)
Not promote			13 (red cards)
	24	24	48

9a. Compute p_{male} = the proportion of supervisors in the male file group who promoted

9b. Compute p_{female} = the proportion of supervisors in the female file group who promoted

9c. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

9d. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 2.

Simulation Rep 3	Male File Group	Female File Group	Total
Promote			35 (blue cards)
Not promote			13 (red cards)
	24	24	48

9e. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

9f. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

9g. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

9h. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 3.

We will analyze your work together to see how unusual the proportion of 0.29 is if the managers really would make the same decision regardless of the gender marked on the file.

After this, we will automate our simulation. You will need to go to this link:

<https://www.lock5stat.com/StatKey/index.html>

Next time, we will learn how to do this “randomization test” in R.

Middle Window Side Table

Is it possible this difference has arisen by random chance alone due to the variability associated with randomly assigning the supervisors to the “male file” and “female file” groups?

We will try to answer this question by considering two competing claims:

Null hypothesis, H_0 : Each supervisor would make the same decision (whether to promote or not) regardless of the gender listed on the file. The difference in promotion proportions of 29.2% between the groups is due to the variability coming from the random assignment of supervisors to the “male file” and “female file” groups.

Alternative hypothesis, H_a : There is at least one supervisor who would decide to promote based on a male file but to not promote based on a female file.

In the study, 35 out of the 48 supervisors decided to promote. One way to examine this situation is to assume that 35 of the supervisors were going to promote regardless of the gender listed on the candidate's file. The key question is **how unlikely is it for the random assignment process alone to randomly place 21 (or more) of these 35 “promoting supervisors” into the “male file” group?**

1. Take a set of 48 playing cards, 35 of which are blue, each representing a supervisor who decides to “promote”, and 13 of which are red, each representing a supervisor who decides to “not promote”. Count the cards to make sure this is what you have.
2. Shuffle the cards well. Then shuffle them some more. And a whole bunch more to make sure they are well mixed.
3. After shuffling well, deal out two groups: 24 of the supervisors to the “male file” group and 24 of the supervisors to the “female file” group.
4. Count how many “promoting supervisors” (blue cards) are in each group, filling out the table below.

Simulation Rep 1	Male File Group	Female File Group	Total
Promote	18	17	35 (blue cards)
Not promote	6	7	13 (red cards)
	24	24	48

5. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$18/24 = .75$$

6. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$17/24 = .7083$$

7. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$.75 - .7083 = .0417$$

8. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 1.

9. Shuffle the cards extremely well again and repeat this process two more times to produce a difference in proportions for two more simulations. Use the tables and questions on this and the next slide to organize your work.

Simulation Rep 2	Male File Group	Female File Group	Total
Promote	17	18	35 (blue cards)
Not promote	7	6	13 (red cards)
	24	24	48

9a. Compute p_{male} = the proportion of supervisors in the male file group who promoted
 $17/24 = .7083$

9b. Compute p_{female} = the proportion of supervisors in the female file group who promoted
 $18/24 = .75$

9c. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$.7083 - .75 =$

9d. Add your results difference to the spreadsheet under your own Simulation Rep 2.

Simulation Rep 3	Male File Group	Female File Group	Total
Promote	19	16	35 (blue cards)
Not promote	5	8	13 (red cards)
	24	24	48

9e. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$19/24 = .792$$

9f. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$16/24 = 0.667$$

9g. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$.792 - .667 = .125$$

9h. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 3.

We will analyze your work together to see how unusual the proportion of 0.29 is if the managers really would make the same decision regardless of the gender marked on the file.

After this, we will automate our simulation. You will need to go to this link:

<https://www.lock5stat.com/StatKey/index.html>

Next time, we will learn how to do this “randomization test” in R.

Middle Door Side Table

Is it possible this difference has arisen by random chance alone due to the variability associated with randomly assigning the supervisors to the “male file” and “female file” groups?

We will try to answer this question by considering two competing claims:

Null hypothesis, H_0 : Each supervisor would make the same decision (whether to promote or not) regardless of the gender listed on the file. The difference in promotion proportions of 29.2% between the groups is due to the variability coming from the random assignment of supervisors to the “male file” and “female file” groups.

Alternative hypothesis, H_a : There is at least one supervisor who would decide to promote based on a male file but to not promote based on a female file.

In the study, 35 out of the 48 supervisors decided to promote. One way to examine this situation is to assume that 35 of the supervisors were going to promote regardless of the gender listed on the candidate's file. The key question is **how unlikely is it for the random assignment process alone to randomly place 21 (or more) of these 35 “promoting supervisors” into the “male file” group?**

1. Take a set of 48 playing cards, 35 of which are blue, each representing a supervisor who decides to “promote”, and 13 of which are red, each representing a supervisor who decides to “not promote”. Count the cards to make sure this is what you have.
2. Shuffle the cards well. Then shuffle them some more. And a whole bunch more to make sure they are well mixed.
3. After shuffling well, deal out two groups: 24 of the supervisors to the “male file” group and 24 of the supervisors to the “female file” group.
4. Count how many “promoting supervisors” (blue cards) are in each group, filling out the table below.

Simulation Rep 1	Male File Group	Female File Group	Total
Promote	17	18	35 (blue cards)
Not promote	7	6	13 (red cards)
	24	24	48

5. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$17/24 = 0.708$$

6. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$18/24 = 0.75$$

7. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.708 - 0.75 = -.042$$

8. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 1.

9. Shuffle the cards extremely well again and repeat this process two more times to produce a difference in proportions for two more simulations. Use the tables and questions on this and the next slide to organize your work.

Simulation Rep 2	Male File Group	Female File Group	Total
Promote	17	18	35 (blue cards)
Not promote	7	6	13 (red cards)
	24	24	48

9a. Compute p_{male} = the proportion of supervisors in the male file group who promoted

$$17/24 = 0.708$$

9b. Compute p_{female} = the proportion of supervisors in the female file group who promoted

$$18/24 = 0.75$$

9c. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.708 - 0.75 = -0.042$$

9d. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 2.

Simulation Rep 3	Male File Group	Female File Group	Total
Promote	18	17	35 (blue cards)
Not promote	6	7	13 (red cards)
	24	24	48

9e. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$18/24 = 0.75$$

9f. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$17/24 = 0.708$$

9g. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.75 - 0.708 = 0.042$$

9h. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 3.

We will analyze your work together to see how unusual the proportion of 0.29 is if the managers really would make the same decision regardless of the gender marked on the file.

After this, we will automate our simulation. You will need to go to this link:

<https://www.lock5stat.com/StatKey/index.html>

Next time, we will learn how to do this “randomization test” in R.

Back Window Side Table

Is it possible this difference has arisen by random chance alone due to the variability associated with randomly assigning the supervisors to the “male file” and “female file” groups?

We will try to answer this question by considering two competing claims:

Null hypothesis, H_0 : Each supervisor would make the same decision (whether to promote or not) regardless of the gender listed on the file. The difference in promotion proportions of 29.2% between the groups is due to the variability coming from the random assignment of supervisors to the “male file” and “female file” groups.

Alternative hypothesis, H_a : There is at least one supervisor who would decide to promote based on a male file but to not promote based on a female file.

In the study, 35 out of the 48 supervisors decided to promote. One way to examine this situation is to assume that 35 of the supervisors were going to promote regardless of the gender listed on the candidate's file. The key question is **how unlikely is it for the random assignment process alone to randomly place 21 (or more) of these 35 “promoting supervisors” into the “male file” group?**

1. Take a set of 48 playing cards, 35 of which are blue, each representing a supervisor who decides to “promote”, and 13 of which are red, each representing a supervisor who decides to “not promote”. Count the cards to make sure this is what you have.
2. Shuffle the cards well. Then shuffle them some more. And a whole bunch more to make sure they are well mixed.
3. After shuffling well, deal out two groups: 24 of the supervisors to the “male file” group and 24 of the supervisors to the “female file” group.
4. Count how many “promoting supervisors” (blue cards) are in each group, filling out the table below.

Simulation Rep 1	Male File Group	Female File Group	Total
Promote	17	18	35 (blue cards)
Not promote	7	6	13 (red cards)
	24	24	48

5. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$17/24=0.708333333333$$

6. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$18/24=.75$$

7. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.708333333333-.75=-0.041666666667$$

8. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 1.

9. Shuffle the cards extremely well again and repeat this process two more times to produce a difference in proportions for two more simulations. Use the tables and questions on this and the next slide to organize your work.

Simulation Rep 2	Male File Group	Female File Group	Total
Promote	16	19	35 (blue cards)
Not promote	8	5	13 (red cards)
	24	24	48

9a. Compute p_{male} = the proportion of supervisors in the male file group who promoted
 $16/24=0.666666666667$

9b. Compute p_{female} = the proportion of supervisors in the female file group who promoted
 $19/24=0.791666666667$

9c. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).
 $0.666666666667 - 0.791666666667 = -.125$

9d. Add the difference to the difference from Simulation Rep 1

Simulation Rep 3	Male File Group	Female File Group	Total
Promote	19	16	35 (blue cards)
Not promote	5	8	13 (red cards)
	24	24	48

9e. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$19/24=0.791666666667$$

9f. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$16/24=0.666666666667$$

9g. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.791666666667 - 0.666666666667 = .125$$

9h. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 3.

We will analyze your work together to see how unusual the proportion of 0.29 is if the managers really would make the same decision regardless of the gender marked on the file.

After this, we will automate our simulation. You will need to go to this link:

<https://www.lock5stat.com/StatKey/index.html>

Next time, we will learn how to do this “randomization test” in R.

Back Door Side Table

Is it possible this difference has arisen by random chance alone due to the variability associated with randomly assigning the supervisors to the “male file” and “female file” groups?

We will try to answer this question by considering two competing claims:

Null hypothesis, H_0 : Each supervisor would make the same decision (whether to promote or not) regardless of the gender listed on the file. The difference in promotion proportions of 29.2% between the groups is due to the variability coming from the random assignment of supervisors to the “male file” and “female file” groups.

Alternative hypothesis, H_a : There is at least one supervisor who would decide to promote based on a male file but to not promote based on a female file.

In the study, 35 out of the 48 supervisors decided to promote. One way to examine this situation is to assume that 35 of the supervisors were going to promote regardless of the gender listed on the candidate's file. The key question is **how unlikely is it for the random assignment process alone to randomly place 21 (or more) of these 35 “promoting supervisors” into the “male file” group?**

1. Take a set of 48 playing cards, 35 of which are blue, each representing a supervisor who decides to “promote”, and 13 of which are red, each representing a supervisor who decides to “not promote”. Count the cards to make sure this is what you have.
2. Shuffle the cards well. Then shuffle them some more. And a whole bunch more to make sure they are well mixed.
3. After shuffling well, deal out two groups: 24 of the supervisors to the “male file” group and 24 of the supervisors to the “female file” group.
4. Count how many “promoting supervisors” (blue cards) are in each group, filling out the table below.

Simulation Rep 1	Male File Group	Female File Group	Total
Promote	17	18	35 (blue cards)
Not promote	7	6	13 (red cards)
	24	24	48

5. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$17/24=0.7083333333$$

6. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$18/24=0.75$$

7. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.7083333333-0.75=-0.0416666667$$

8. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 1.

9. Shuffle the cards extremely well again and repeat this process two more times to produce a difference in proportions for two more simulations. Use the tables and questions on this and the next slide to organize your work.

Simulation Rep 2	Male File Group	Female File Group	Total
Promote	18	17	35 (blue cards)
Not promote	6	7	13 (red cards)
	24	24	48

9a. Compute p_{male} = the proportion of supervisors in the male file group who promoted
 $18/24=0.75$

9b. Compute p_{female} = the proportion of supervisors in the female file group who promoted
 $17/24=0.7083333333$

9c. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$0.75-0.7083333333=0.0416666667$

Simulation Rep 3	Male File Group	Female File Group	Total
Promote	17	18	35 (blue cards)
Not promote	7	6	13 (red cards)
	24	24	48

9e. Compute p_{male} = the proportion of supervisors in the male file group who promoted (show work)

$$17/24=0.7083333333$$

9f. Compute p_{female} = the proportion of supervisors in the female file group who promoted (show work)

$$18/24=0.75$$

9g. Compute the difference: $p_{\text{male}} - p_{\text{female}}$ (show work).

$$0.7083333333-0.75=-0.0416666667$$

9h. Add your group's difference to the [spreadsheet](#) under your group's Simulation Rep 3.

We will analyze your work together to see how unusual the proportion of 0.29 is if the managers really would make the same decision regardless of the gender marked on the file.

After this, we will automate our simulation. You will need to go to this link: [StatKey](#)

Next time, we will learn how to do this “randomization test” in R.